THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 9, 4, 10, 13, 4 respectively, The mode of the data is: 1

(a) 70 **(b)** 80 **(c)** 64.75 **(d)** 72.5

2. A letter is chosen at random from the word 'MATHEMATICS'. The probability that it is a vowel is: 1

(a) $\frac{3}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{5}{11}$ **(d)** $\frac{7}{11}$

3. The decimal expansion of $\frac{13}{3125}$ terminates after how many places of decimals? 1

(a) 6 **(b)** 8 **(c)** 5 **(d)** 7

4. If α, β are zeroes of $x^2 - 9x + 14$, then $\alpha + \beta$ is: 1

(a) 10 **(b)** 9 **(c)** 14 **(d)** -9

5. For the pair $x + y = 10$ and $x - y = 2$, the value of xy is: 1

(a) 48 **(b)** 10 **(c)** 24 **(d)** 26

6. The roots of $x^2 - 8x + 15 = 0$ are: 1

(a) 3 and -5 **(b)** -3 and -5 **(c)** -3 and 5 **(d)** 3 and 5

7. If $S_n = n^2 + 2n$, then $a_1 =$ 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 1

8. A man goes 15 m due east and then 20 m due north. His distance from the start (m) is: 1

(a) 625 **(b)** 5 **(c)** 35 **(d)** 25

9. If the midpoint of $(2, 3)$ and (x, y) is $(5, 6)$, then $x =$ 1

(a) 8 **(b)** $\frac{7}{2}$ **(c)** 3 **(d)** 12

10. $\tan 20^\circ \cdot \tan 70^\circ =$ 1

(a) 20 **(b)** 1 **(c)** 0 **(d)** 2

11. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 20 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. The height of the tower is: 1

(a) 10 m **(b)** $20\sqrt{3}$ m **(c)** $\frac{20\sqrt{3}}{3}$ m **(d)** $10\sqrt{3}$ m

12. PA and PB are tangents from an external point P to a circle with centre O, touching it at A and B. $\angle APB = 60°$. The $\angle OAB$ is: 1

(a) $90°$ **(b)** $30°$ **(c)** $60°$ **(d)** $120°$

13. The radii of two circles are 6 cm and 8 cm. The radius of the circle whose area equals the sum of their areas is: 1

(a) 2 cm **(b)** 12 cm **(c)** 10 cm **(d)** 14 cm

14. From a solid cylinder of height 24 cm and base radius 7 cm, a conical cavity of the same height and the same base radius is hollowed out. The total surface area of the remaining solid is: 1

(a) 1606 cm² **(b)** 1760 cm² **(c)** 1914 cm² **(d)** 1364 cm²

15. For the data with classes 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 3, 8, 20, 12, 7 respectively, The modal class is: 1

(a) 40–50 **(b)** 10–20 **(c)** 30–40 **(d)** 20–30

16. A bag contains 4 red and 6 blue balls. 2 more red balls are put into the bag and a ball is drawn at random. The probability that the ball drawn is red is: 1

(a) $\frac{5}{12}$ **(b)** $\frac{7}{12}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{6}{13}$

17. Which of the following is irrational? 1

(a) $\sqrt{5}+3$ **(b)** $\sqrt{9}+3$ **(c)** $\sqrt{25}-3$ **(d)** $3\times\sqrt{9}$

18. If α, β are zeroes of $x^2 - 9x + 14$, then $\alpha\beta$ is: 1

(a) -14 **(b)** 9 **(c)** 15 **(d)** 14

19. The pair $x + 2y = 3$ and $2x + ky = 7$ has no solution when $k =$ 1

(a) 2 **(b)** 6 **(c)** 5 **(d)** 4

20. The discriminant of $2x^2 - 7x + 3 = 0$ is: 1

(a) 73 **(b)** 25 **(c)** 37 **(d)** -25

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. The marks obtained by students in a test are given below. 2

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	20	22	20	17	6	15

Find the mean.

22. A coin is tossed three times. Find the probability of getting at least one tail. 2

23. Given that HCF(306,657)=9, find LCM(306,657). 2

24. Find the sum and product of the zeroes of $2x^2 - 8x + 6$. 2

25. Solve by substitution: $x + y = 8$, $x - y = 2$. 2

26. Find the roots of $2x^2 - 7x + 6 = 0$ by factorisation. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Solve by elimination method: $2x + 3y = 12$ and $5x + 4y = 23$. 3

28. Find the roots of $x^2 - 6x + 8 = 0$ using the quadratic formula (if they exist). 3

29. How many terms are there in the AP $18, \frac{31}{2}, 13, \ldots, -\frac{19}{2}$? 3

30. The areas of two similar triangles are 16 cm² and 36 cm². If a side of the first is 4 cm, find the corresponding side of the second. 3

31. Find the area of the triangle with vertices $(1, 1)$, $(4, 5)$ and $(7, 2)$. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 7$ cm and $BC = 24$ cm. Find $\sin A$, $\cos A$ and $\tan A$. 3

32. From the top of a building 5 m high, the angle of elevation of the top of a tower is 60° and the angle of depression of its foot is 45°. Find the height of the tower. 3

OR

In $\triangle ABC$, right-angled at B, $AB = 20$ cm and $BC = 21$ cm. A circle is inscribed in the triangle. Find the radius of the inscribed circle. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A loan is repaid in 20 monthly instalments; the first is ₹100 and each later instalment is ₹10 more than the previous. Find the instalment in the last month and the total amount repaid. 5

34. Two similar triangles have perimeters 18 cm and 36 cm. If a side of the smaller is 5 cm, find the corresponding side of the larger and the ratio of their areas. 5

35. Three vertices of a parallelogram $ABCD$ taken in order are $A(1, 2)$, $B(4, 3)$ and $C(5, 7)$. Find the fourth vertex D. 5

36. If $\tan A = \frac{4}{3}$, evaluate $\dfrac{4\sin A + 3\cos A}{4\sin A + 3\cos A}$. 5

OR

A statue 2 m tall stands on a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. Find the height of the pedestal. 5

37. In $\triangle ABC$, $\angle B = 90^\circ$, $AB = 6$ cm and $BC = 8$ cm. A circle touching all three sides touches AB at F and BC at D. Find the radius of the circle and the lengths AF and CD (in this order). 5

OR

A brooch is a circular piece of silver wire of diameter 35 mm. The wire is also used for 5 equal diameters that divide the brooch into 10 sectors. Find the total length of wire required (in mm). 5

38. A conical vessel of radius 6 cm and height 8 cm is full of water. The water is poured into a cylindrical vessel of base radius 4 cm. Find the height of the water in the cylindrical vessel. 5

OR

The median of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60, 60–70 and frequencies 15, 3, x, 16, y, 10, 13 respectively, and total frequency 100, is 36.25. Find the value of x. 5

END OF QUESTION PAPER — Total: 80 Marks